ALGEBRA 1 · UNIT 3: LINEAR INEQUALITIES

Class Trip Planner

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students plan a class field trip under real constraints: a fixed budget, bus capacity, group-rate minimums, ride temperature limits, and lunch money. They write, solve, and graph single and compound inequalities, then propose their own trip.

Students see that real decisions are limited by several constraints at once, and that the answer to an inequality must be interpreted (whole students, rounding down for budgets and up for minimums). Day 3 lets students choose a destination and build their own proposal.

Standards Addressed

- A-CED.1 Create inequalities in one variable and use them to solve problems
- A-CED.3 Represent constraints by inequalities and interpret solutions as viable or nonviable
- A-REI.3 Solve linear inequalities in one variable
- A-SSE.1 Interpret expressions in context (Honors: absolute value)

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch: How many students could go on a $2,000 budget? Collect guesses. Part 1: budget inequality and number line.
Day 2	Part 2: compound inequalities (and or). Use the temperature rule to discuss 'and' vs 'or'.
Day 3	Part 3: choose a destination, write all constraints, and build the proposal. Short presentations or gallery walk.

Materials

- Student handout
- Calculators
- Colored pencils for number lines
- Poster paper or slide template

How the Three Versions Differ

Level	What changes
CP2	One destination, a guided budget inequality, number lines provided, and one compound 'and' inequality with support.
CP1	Adds fundraising, error analysis with flipping the sign, a Celsius conversion inside a compound inequality, and an 'or' inequality.
Honors	Generalizes the budget formula, adds a second bus (piecewise reasoning), absolute value inequalities, and a discount that changes which constraint is binding.

Watch For (Common Misconceptions)

- Rounding up a budget answer (55.4 students does not mean 56 can go).
- Forgetting to flip the inequality when dividing by a negative.
- Mixing up 'and' (between) with 'or' (outside).

Extension Ideas

- Use real prices from a local destination.
- Add chaperones (1 per 10 students) whose tickets also come from the budget.
- Graph cost vs. students and show the budget as a horizontal line (previews Unit 5).

Answer Key – CP2

Part 1: Stay Within the Budget

#	Task	Answer
1	Write an expression for the total cost of the trip if n students go.	450 + 28n
2	The class can spend at most $2,000. Write an inequality using your expression.	450 + 28n ≤ 2000
3	Solve your inequality. What is the greatest number of students who can go? Explain why you round down.	28n ≤ 1550 → n ≤ 55.36 → at most 55 students (you cannot take part of a student, and 56 is over budget).
4	Show your solution on the number line.	Closed dot at 0 and at 55 with whole numbers shaded between them (or shade to 55).
5	Check: find the total cost for 55 students and for 56 students. Which one is over budget?	55 students: 450 + 1540 = $1,990 (OK). 56 students: 450 + 1568 = $2,018 (over budget).

Part 2: Follow the Rules

#	Task	Answer
6	The group rate needs at least 20 students. Combine this with your budget answer to write a compound inequality for n.	20 ≤ n ≤ 55
7	Rides are open only from 50°F to 95°F. Write a compound inequality for the temperature T when rides are open, and graph it.	50 ≤ T ≤ 95; closed dots at 50 and 95, shaded between.
8	Write an inequality for the temperatures when rides are closed.	T < 50 or T > 95
9	Each student has $15 for lunch. The meal costs $8.50 and souvenirs cost $2.50 each. Write and solve an inequality for the number of souvenirs s a…	8.50 + 2.50s ≤ 15 → 2.50s ≤ 6.50 → s ≤ 2.6 → at most 2 souvenirs

Part 3: Plan Your Own Trip

#	Task	Answer
10	Which destination did you choose? Write an inequality for the number of students who can go, and solve it.	Aquarium: $450 + 22n \le 2000 \rightarrow n \le 70.45 \rightarrow 70$ (but the bus holds 56, so 56). Museum (with planetarium): $450 + 24n \le 2000 \rightarrow n \le 64.58 \rightarrow 64$ (bus limit 56). Museum without planetarium: $450 + 18n \le 2000 \rightarrow n \le 86.1$ (bus limit 56). Adventure Park: $n \le 55$.
11	Write a compound inequality that includes the group-rate rule (at least 20) and your budget answer.	Aquarium or Museum: $20 \le n \le 56$ (bus limit). Adventure Park: $20 \le n \le 55$.
12	Write 2–3 sentences to the principal explaining why your trip should be approved.	Answers vary.

Scoring note: On Day 3, praise students who notice the bus holds only 56 students, which is a second constraint.

Answer Key – CP1

Part 1: Budget & Fundraising

#	Task	Answer
1	Write and solve an inequality to find the greatest number of students who can go on a $2,000 budget.	450 + 28n ≤ 2000 → n ≤ 55.36 → at most 55 students
2	The class account actually has only $1,200 right now. The class will sell snacks for $1.50 profit each. How many snacks must be sold so that 45…	Cost for 45: 450 + 28(45) = $1,710. 1200 + 1.50x ≥ 1710 → 1.50x ≥ 510 → x ≥ 340 snacks
3	Riley solved 2000 − 28n ≥ 450 and wrote: −28n ≥ −1550, so n ≥ 55.36. Explain the mistake and give the correct solution.	Dividing by −28 must flip the sign: n ≤ 55.36, so at most 55 students.
4	Show the correct solution for the number of students on the number line.	Closed dot at 55, shaded left to 0 (whole numbers only).

Part 2: Compound Conditions

#	Task	Answer
5	Write a compound inequality for the number of students n that satisfies the group-rate rule, the budget, and the bus size.	20 ≤ n ≤ 55 (the budget limit, 55, is less than the bus limit, 56)
6	The weather app shows Celsius. Rides are open when 50 ≤ F ≤ 95, and F = 1.8C + 32. Write and solve a compound inequality to find the Celsius…	50 ≤ 1.8C + 32 ≤ 95 → 18 ≤ 1.8C ≤ 63 → 10 ≤ C ≤ 35
7	The trip includes 2 hours of bus travel plus h hours at the park. The bus company adds a fee if the trip is shorter than 4 hours or longer than 10…	2 + h < 4 or 2 + h > 10 → h < 2 or h > 8; open dots at 2 and 8, shaded outward
8	Write and solve an inequality for the number of souvenirs a student can buy with $15 if lunch costs $8.50 and souvenirs cost $2.50 each.	8.50 + 2.50s ≤ 15 → s ≤ 2.6 → at most 2 souvenirs

Part 3: Plan Your Own Trip

#	Task	Answer
9	Write and solve an inequality for the greatest number of students who can go. Remember the bus limit.	Aquarium: $n \le 70.45$, bus limit gives $n \le 56$. Museum with planetarium: $n \le 64.58$, bus limit 56. Adventure Park: $n \le 55$.
10	Choose a realistic group size. Write and solve an inequality to find how many snacks must be sold to pay for that group.	Answers vary. Example: Aquarium with 50 students costs $450 + 22(50) = \$1{,}550$; $1200 + 1.50x \ge 1550 \rightarrow x \ge 233.3 \rightarrow$ 234 snacks.
11	Write at least two more constraints for your trip (time, temperature, lunch, chaperones) as inequalities and solve them. Which constraint limits…	Answers vary.

Answer Key – Honors

Part 1: Budget, Buses & Fundraising

#	Task	Answer
1	Let B = bus cost, p = ticket price, and M = budget. Write an inequality for the number of students n, and solve it for n.	$B + pn \le M \rightarrow n \le (M - B) \div p$
2	Use your formula to find the most students that can go with one bus on a $2,000 budget.	$n \le (2000 - 450) \div 28 = 55.36 \rightarrow$ 55 students
3	The principal offers $3,000 if the whole grade goes, which needs two buses. Write and solve an inequality for n. Then write a compound inequality…	$900 + 28n \le 3000 \rightarrow 28n \le 2100 \rightarrow n \le 75$. Two buses are needed when $n \ge 57$, so $57 \le n \le 75$.
4	The class has $1,200 saved and earns $1.50 profit per snack. How many snacks must be sold to take 70 students (two buses)?	Cost = 900 + 28(70) = $2,860. $1200 + 1.50x \ge 2860 \rightarrow 1.50x \ge 1660 \rightarrow x \ge 1106.7 \rightarrow$ 1,107 snacks

Part 2: Ranges & Absolute Value

#	Task	Answer
5	Rides open when $50 \le F \le 95$. Using $F = 1.8C + 32$, find the open range in Celsius.	$18 \le 1.8C \le 63 \rightarrow 10 \le C \le 35$
6	Write the Fahrenheit range $50 \le F \le 95$ as one absolute value inequality. Explain the meaning of each number.	$\lvert F - 72.5 \rvert \le 22.5$. The center 72.5 is the middle of the range; 22.5 is the distance from the center to each end.
7	The trip is 2 hours of bus travel plus h hours at the park. A fee is charged if the total is under 4 hours or over 10 hours. Solve for h, then…	$2 + h < 4$ or $2 + h > 10 \rightarrow h < 2$ or $h > 8 \rightarrow \lvert h - 5 \rvert > 3$
8	The park gives a 15% ticket discount to groups of 40 or more. With one bus and $2,000, find every group size that is affordable (group rate…	Under 40: $450 + 28n \le 2000 \rightarrow n \le 55$, so $20 \le n \le 39$ all work. 40 or more: $450 + 23.80n \le 2000 \rightarrow n \le 65.1$, but the bus holds 56, so $40 \le n \le 56$. Affordable: $20 \le n \le 56$. The bus capacity is now the binding constraint.

Part 3: Compare & Recommend

#	Task	Answer
9	For each of two destinations, write and solve the inequalities for group size and buses needed. Which destinations can take the whole grade?	Answers vary. Example: Aquarium, 110 students, 2 buses: 900 + 22(110) = $3,320 > $3,000, so not the whole grade ($n \le 95.45 \rightarrow 95$). Museum without planetarium: 900 + 18(110) = $2,880 ≤ $3,000, so the whole grade can go. Adventure Park: $900 + 28n \le 3000 \rightarrow n \le 75$.
10	For your chosen destination, write the fundraising inequality and find the snacks needed for the largest affordable group.	Answers vary. Museum (no planetarium), 110 students: $1200 + 1.50x \ge 2880 \rightarrow x \ge 1120$ snacks.
11	Write a recommendation that names the binding constraint (the one that limits the plan) and explains how the plan would change if that constraint…	Answers vary.

Scoring note: Students may treat the planetarium as optional. Accept either choice if it is stated clearly.

Scoring Rubric (all levels)

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Every constraint is written and solved correctly, and the final group size satisfies all of them. Goes beyond the minimum with a creative, realistic choice.	Every constraint is written and solved correctly, and the final group size satisfies all of them.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Proposal shows the destination, every constraint as an inequality, graphs, and a final recommendation. Polished, organized, and visually engaging.	Proposal shows the destination, every constraint as an inequality, graphs, and a final recommendation. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.